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RESEARCH ARTICLE

Excitation-Contraction Coupling, Electrophysiology, and Arrhythmias

hiPSC-CM electrophysiology: impact of temporal changes and study parameters on experimental reproducibility

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Abstract

Human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) are frequently used for preclinical cardiotoxicity testing and remain an important tool for confirming model-based predictions of drug effects in accordance with the comprehensive in vitro proarrhythmia assay (CiPA). Despite the considerable benefits hiPSC-CMs provide, concerns surrounding experimental reproducibility have emerged. We investigated the effects of temporal changes and experimental parameters on hiPSC-CM electrophysiology. iCell cardiomyocytes^2 were cultured and biosignals were acquired using a microelectrode array (MEA) system (2–14 days). Continuous recordings revealed a 22.6% increase in the beating rate and 7.7% decrease in the field potential duration (FPD) during a 20-min equilibration period. Location-specific differences across a multiwell plate were also observed, with iCell cardiomyocytes^2 in the outer rows beating 8.8 beats/min faster than the inner rows. Cardiac endpoints were also impacted by cell culture duration; from 2 to 14 days, the beating rate decreased (−12.7 beats/min), FPD lengthened (+257 ms), and spike amplitude increased (+3.3 mV). Cell culture duration (4–10 days) also impacted cardiomyocyte drug responsiveness (E-4031, nifedipine, isoproterenol). qRT-PCR results suggest that daily variations in cardiac metrics may be linked to the continued maturation of hiPSC-CMs in culture (2–30 days). Daily experiments were also repeated using a second cell line (Cor.4U). Collectively, our study highlights multiple sources of variability to consider and address when performing hiPSC-CM MEA studies. To improve reproducibility and data interpretation, MEA-based studies should establish a standardized protocol and report key experimental conditions (e.g., cell line, culture time, equilibration time, electrical stimulation settings, and raw data values).

NEW & NOTEWORTHY We demonstrate that iCell cardiomyocytes^2 electrophysiology measurements are impacted by deviations in experimental techniques including electrical stimulation protocols, equilibration time, well-to-well variability, and length of hiPSC-CM culture. Furthermore, our results indicate that hiPSC-CM drug responsiveness changes within the first 2 wk following defrost.

CIPA; electrophysiology; experimental reproducibility; hiPSC-CMs; microelectrode array

INTRODUCTION

Human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) have been widely adopted as a tool for translational research studies focused on the efficacy, safety, and cardiotoxicity of drug compounds and environmental chemicals. As previously reviewed (1–4), hiPSC-CMs offer several advantages to the cardiac research field: *1*) hiPSC-CMs are a renewable source of human-derived tissue; *2*) hiPSC-CMs are sensitive and responsive to drugs/chemicals that cause cardiotoxicity through structural, electrophysiology, or contractile disturbances; *3*) patient-derived hiPSC-CMs can be generated for disease modeling and precision medicine; and *4*) hiPSC-CMs more accurately recapitulate human cardiac biology, as compared with other animal models. Although animal models may fail to predict drug safety or efficacy because of species-specific differences (5, 6), hiPSC-CMs express similar cardiac ion channels as human cardiomyocytes, have an action potential duration that is comparable with the human electrocardiographic QT interval, and hiPSC-CMs are highly responsive to ion channel blocking drugs (7–13). One

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limitation to using a hiPSC-CM model is the immature phenotype of these cells [e.g., small size, spontaneous beating activity (14, 15)]; although, multiple technical strategies have been identified to effectively mature these cells to more closely mimic adult human cardiomyocytes (16–21).

Given these advantages, hiPSC-CMs have been incorporated as an integral tool in the Comprehensive In Vitro Proarrhythmia (CiPA) initiative, which was established to more precisely identify drug compounds with proarrhythmic risk using nonclinical assays (22, 23). Indeed, hiPSC-CMs were shown to accurately predict the proarrhythmic potential of ~85% of drugs (26 tested) known to cause clinical QT prolongation (24). Furthermore, hiPSC-CMs were both sensitive (87%) and specific (70%) when used to screen 51 drug compounds for their effects on cardiac contraction (25). Such large-scale cardiotoxicity screening assays can be performed using a high-throughput microelectrode array (MEA) platform, which allows one to measure both electrical and contractile properties of hiPSC-CMs. Briefly, hiPSC-CMs are plated atop microelectrodes embedded within a cell culture plate, which allows for continuous (noninvasive) recordings of *1*) extracellular field potentials akin to an electrocardiogram, *2*) local extracellular action potentials akin to a monophasic action potential, and *3*) impedance signals corresponding to cardiomyocyte contractility (26–28). Although hiPSC-CM MEA studies can be used to identify cardiotoxic compounds, multiple studies have reported some inconsistences in the acquired data across study sites and cell lines (24, 29–35). Multiple factors are likely to contribute to such data discrepancies, including phenotypic differences in each individual hiPSC-CM line used, cell culture conditions (e.g., plating density, duration of cell culture, serum vs. serum-free media), and the recording protocol implemented (e.g., gas/temperature control, length of equilibration time, pacing parameters). For example, the hiPSC-CM spontaneous beating rate is influenced by both the recording temperature and plating density (36–38), and cells with a slower beating rate can have an exaggerated proarrhythmic response to I_{Kr} blockers (31). With this in mind, we aimed to investigate experimental factors that may influence hiPSC-CM MEA study results. Clearly defining study parameters that can influence MEA results will help to improve data interpretation and experimental reproducibility between laboratories, which is an increasing focus of the cardiovascular research field (39).

METHODS

Culture and Maintenance of iCell Cardiomyocytes^2 for MEA Studies

MEA electrodes (Cat. No. M384-tMEA-24W and Cat. No. M384-BIO-24; Axion Biosystems) were coated with 50 μg/mL fibronectin (Cat. No. 33010018; Gibco) diluted in sterile phosphate-buffered saline (Cat. No. 10010023; Gibco) for 1 h in a cell culture incubator (37°C, 5% CO_2). Subsequently, fibronectin was removed, and iCell cardiomyocytes^2 (Cat. No. 01434; Fujifilm Cellular Dynamics) were plated atop a 16-electrode array at a density of 40–65,000 cells/well in iCell cardiomyocyte plating media (Cat. No. M1001; Fujifilm Cellular Dynamics). hiPSC-CMs were maintained in a cell culture incubator for 1–3 h to allow cells to adhere to the electrode array, and thereafter, the media was removed and replaced with iCell serum-based maintenance media (Cat. No. M1003; Fujifilm Cellular Dynamics) supplemented with 2 μg/mL ciprofloxacin (Cat. No. 17850-5 G-F; Sigma-Aldrich). A full volume media change occurred 1 day post-defrost; half volume media changes occurred every 2–3 days for the remainder of the experimental period.

Microelectrode Array Recordings

Equilibration.

A Maestro Edge MEA (Axion BioSystems) was used for all hiPSC-CM physiological measurements; all experiments were conducted at 37°C with 5% CO_2. To assess phenotypic changes that may occur during equilibration to the MEA system, cellular recordings were acquired continuously for 20 min while hiPSC-CMs were beating spontaneously (*day 4* post-defrost). For all subsequent experiments, hiPSC-CMs were allowed to equilibrate to the MEA system for 20 min to reach steady state conditions. Thereafter, cellular recordings were acquired while hiPSC-CMs were beating spontaneously and in response to external pacing using the Axion Biosystems "cardiac pacing stimulation" protocol.

MEA parameters of interest.

Biosignals were acquired from hiPSC-CMs plated atop a 16-electrode array (Fig. 1). Electrophysiology parameters of interest included the spontaneous beating rate, field potential duration (FPD), spike amplitude, and conduction velocity across the electrode array (Fig. 1*C*). As previously described, the spike amplitude is an indicator of depolarization (akin to an electrocardiogram QRS complex), and FPD is an indicator of repolarization (akin to QT interval) (26). Contractility parameters of interest included excitation contraction delay and contraction beat amplitude (Fig. 1*D*). Excitation contraction delay quantifies the time between the depolarization spike and the contractility waveform peak, while the beat amplitude is measured by a change in impedance as hiPSC-CMs contract and relax (40, 41). In a subset of experiments, cardiomyocyte-electrode coupling was enhanced to record local extracellular action potentials (LEAP) (28) to calculate the action potential duration (APD) at 30% (APD_{30}), 50% (APD_{50}), and 90% (APD_{90}) repolarization (Fig. 1*E*). Data points from individual wells were excluded if hiPSC-CMs were not beating at the desired pacing frequency, or if the signal-to-noise ratio was too small for accurate measurements. Because of physical and electrical coupling between cardiomyocytes, it is common practice to select a single representative electrode for signal analysis (26, 32, 42). See Supplemental Fig. S1 (all Supplemental figures and tables are available at https://doi.org/10.6084/m9.figshare.24222883) for validation of FPD signal homogeneity across multiple electrodes (1.85–3.79% coefficient of variation).

Daily hiPSC-CM maturation.

We tested the impact of cell culture duration on hiPSC-CM physiology by acquiring biosignals each day during the first 2 wk after defrost (Fig. 1*A*). Field potential recordings and contractility measurements were acquired from spontaneously

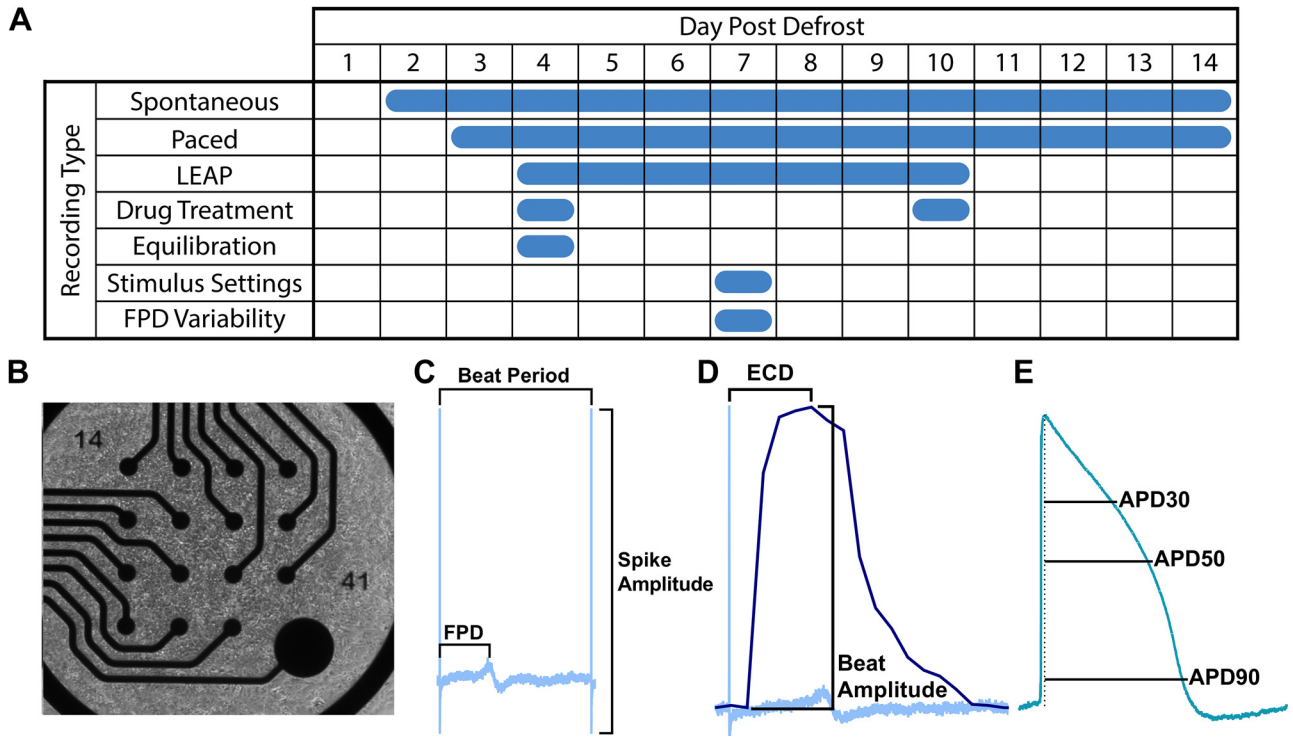


Figure 1. Experimental design. *A*: time line of iCell Cardiomyocytes[2] microelectrode array (MEA) recordings. *B*: example image of human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) plated onto an MEA. *C*: representative extracellular field potential waveform, with markers indicating the beat period, field potential duration (FPD), and spike amplitude. *D*: representative contraction recording, with markers indicating the excitation-contraction delay (ECD) and beat amplitude. *E*: representative local extracellular action potential (LEAP) recording with markers indicating action potential duration (APD) at 30% (APD_{30}), 50% (APD_{50}), and 90% (APD_{90}) repolarization.

beating hiPSC-CMs 2 to 14 days post-defrost; these signals were also recorded in response to external pacing (3 to 14 days post-defrost) after cells were fully attached and formed a confluent monolayer. LEAP recordings were also acquired in response to external pacing (4 to 10 days post-defrost). For paced recordings, hiPSC-CMs were stimulated at 1.5 Hz (stimulus settings, 400 μs, 800 mV, and 20 μA).

Electrical stimulation settings.

We tested the impact of adjusting stimulation settings on cardiac electrophysiology endpoints. To negate the impact of cell culture time, these experiments were only performed on *day 7* post-defrost. Variations in current (20–50 μA), stimulus duration (400–1,000 μs), and voltage (800–1,200 mV) were tested. Only one parameter was adjusted at a time; otherwise, the settings were held constant (stimulus duration, 400 μs; voltage, 800 mV; current, 20 μA).

Drug response.

To measure the impact of cell culture time on hiPSC-CM drug responsiveness, cells were exposed to three different drugs at two different time points (*day 4*, *day 10* post-defrost; Fig. 1*A*). A hERG channel blocker (50 nM E-4031, Cat. No. 15203, Cayman Chemical) was tested for its ability to prolong FPD and APD (28, 43, 44). E-4031 reportedly suppresses spontaneous activity at concentrations above 30 nM (45). A calcium channel blocker (100 nM nifedipine, Cat. No. N7634-1G Sigma-Aldrich) was tested for its ability to shorten APD (28) and increase the spontaneous beating rate (44). A β-adrenergic agonist (100 nM isoproterenol, Cat. No. I5627-25G; Sigma-Aldrich) was tested for its ability to shorten APD, increase the spontaneous beating rate, and increase conduction velocity (45, 46). Since these pharmacological agents can alter the spontaneous beating rate, electrical signals were recorded in response to external pacing (1.5 Hz for E-4031, 3 Hz for nifedipine and isoproterenol). Stimulus settings were the same for all drug exposure studies (stimulus duration, 400 μs; voltage, 800 mV; current, 20 μA). Drug treatments were applied for 15 min before recording electrical signals.

Real-Time Quantitative Reverse Transcription-PCR

Total RNA was isolated from hiPSC-CMs on *days 2*, *7*, *14*, and *30* post-defrost using a RNeasy micro kit (Qiagen). RNA concentration was determined using a NanoDrop spectrophotometer (Thermo Fisher), and then RNA was reverse transcribed using a SuperScript VILO cDNA synthesis kit (Invitrogen). TaqMan gene expression assays were used for fluorescence-based real-time quantitative reverse transcription-PCR (qRT-PCR via a QuantStudio 7 platform (Applied Biosystems). Gene expression was normalized to a reference gene glyceraldehyde-3-phosphate dehydrogenase (*GAPDH*) and fold change values are reported (average of $n = 3$ technical replicates) for each assay ($n = 3$–5 individual biological replicates).

Validation of Key Results Using an Alternative hiPSC-CM Cell Line

Cor.4U cardiomyocytes (Cat. No. AX-B-HC02, Axiogenesis/Ncardia) were defrosted and cultured using the same protocol as described for iCell cardiomyocytes^2, with the exception that Cor.4U cells were maintained using Cor.4U complete culture medium (Cat. No. Ax-M-HC250, Axiogenesis/Ncardia). Validation studies included daily measurements of FPD during spontaneous beating (2–14 days post-defrost) and in response to external stimulation at 2 Hz (3–14 days post-defrost) (Supplemental Fig. S2). Note that a slightly faster pacing frequency was used because Cor.4U cells have a faster intrinsic beating rate, as compared with iCell cardiomyocytes^2. We also performed equilibration measurements (*day 7* post-defrost) and assessed gene expression (*days 2*, *7*, *14*, and *30* post-defrost) as described for iCell cardiomyocytes^2.

Data Acquisition and Statistical Analysis

For each measurement, biosignals were collected for 30 s using an MEA system (12.5 kHz sampling frequency). Each MEA file was then manually reviewed (D.G., J.P.) to select one electrode per well with the best signal quality (e.g., action potential or t-wave). Whenever possible, signals were selected for FPD analysis that had an upright t-wave or the positive portion of a biphasic t-wave (see Fig. 1 and Supplemental Fig. S1). After manually selecting one electrode per well, waveforms from that recording electrode were averaged (30 s). Individual wells were excluded if the signal amplitude was insufficient for accurate measurements, or when hiPSC-CMs were not beating at the desired pacing frequency. Outliers were identified and removed using the ROUT method (5% false discovery rate). Results are reported throughout as means ± SD. Statistical analysis was performed using either a Student's *t* test (two groups), one- or two-way analysis of variance (ANOVA) with Holm–Sidak multiple comparisons test (three or more groups), as indicated in each figure legend. An adjusted P value < 0.05 was considered statistically significant. Data are graphically represented as the mean with range (minimum to maximum), unless otherwise indicated. Replicates are defined in each figure.

RESULTS

MEA Equilibration Time Impacts hiPSC-CM Electrophysiology Measurements

For our initial studies, we first used a commercially available hiPSC-CM cell line (iCell Cardiomyocytes^2, Cat. No. 01434, Fujifilm Cellular Dynamics) that has been widely used in preclinical cardiac safety pharmacology studies (27, 30–32, 47–50). The manufacturer's instructions (51) recommend equilibrating hiPSC-CMs in an MEA system for ~10 min before recording electrical signals, although some studies have reported using a longer equilibration time (≥15–20 min) (44, 52–55). To determine whether the equilibration period influenced experimental results, we continuously recorded electrical signals after transferring iCell Cardiomyocytes^2 culture plates to an MEA system. Over the course of a 20-min recording, the spontaneous beating rate increased markedly by 22.6% (2 min, 40.2 ± 2.7; 20 min, 49.3 ± 5.9 beats/min; 1.98 Cohen effect size; $P < 0.0001$; Fig. 2*A*) and the FPD shortened by 7.7% (2 min, 385.8 ± 19.8, 20 min, 356.1 ± 18.8 ms; 1.54 Cohen effect size; $P < 0.0001$; Fig. 2, *C* and *D*). Adaptations in the spontaneous beating rate and FPD were most exaggerated at the beginning of the recording, reaching a plateau near the end of the equilibration period. For example, compared with the last measurement at 20 min, the spontaneous beating rate was significantly slower only between 2 and 12 min ($P < 0.05$) and FPD was longer only between 2 and 7 min ($P < 0.05$). FPD plateaued as the cells reached a steady state, with a mean difference of less than 6 ms for each measurement between 8 and 20 min. No significant differences were observed in the depolarization spike amplitude or conduction velocity during the equilibration period (Fig. 2, *E–H*). To reduce hiPSC-CM equilibration as a confounding factor, it is recommended that hiPSC-CM acclimate to the MEA system for 15–20 min before performing subsequent experiments. For reference, measurements (in 60-s intervals) are available in Supplemental Table S1.

Unexpectedly, we identified location-specific differences in iCell Cardiomyocytes^2 electrophysiology measurements according to the well location on the MEA plate (Fig. 2, *A–D*). Throughout the 20-min equilibration period, we noted a significantly faster beating rate in the outer rows ('A' and 'D') versus the inner rows ('B' and 'C'), which became exaggerated over time. At the 2-min recording time, a modest 4.6% difference was observed between outer (41.1 ± 2.3 beats/min) and inner rows (39.2 ± 2.8 beats/min, 0.74 Cohen effect size, $P < 0.001$), which increased to a 16.4% difference at the 20-min recording time (outer rows, 53.7 ± 4.5; inner rows, 44.9 ± 3.4 beats/min; 2.21 Cohen effect size; $P < 0.0001$; Fig. 2*A*). FPD measurements also differed by well location; at the 20-min recording time, iCell Cardiomyocytes^2 plated in the outer rows had a shorter FPD (351.2 ± 18.0 ms) compared with cells in the inner rows (360.9 ± 18.5 ms, 0.53 Cohen effect size, $P < 0.05$; Fig. 2*C*). No significant difference in spike amplitude or conduction velocity was observed between the inner and outer rows (Fig. 2, *E* and *G*). To reduce well location as a confounding factor, it is recommended that hiPSC-CM treatment groups are dispersed across the MEA multiwell plate, rather than grouping a specific treatment group across a single row.

Electrical Stimulation Settings Influence hiPSC-CM Electrophysiology Measurements

Empirically, we noted that certain pharmacological agents altered the excitability of hiPSC-CMs, preventing the cell monolayer from capturing at a desired pacing frequency. This finding has been reported by others, wherein hiPSC-CMs continue to beat spontaneously at one-half or one-third the applied pacing rate (56). To remedy this problem, it can be helpful to adjust the electrical stimulation settings (i.e., pulse duration, voltage amplitude, and/or current) to successfully pace hiPSC-CMs at a desired frequency. We measured iCell Cardiomyocytes^2 electrophysiology endpoints in response to standard electrical stimulations settings, and then modified each variable independently to quantify

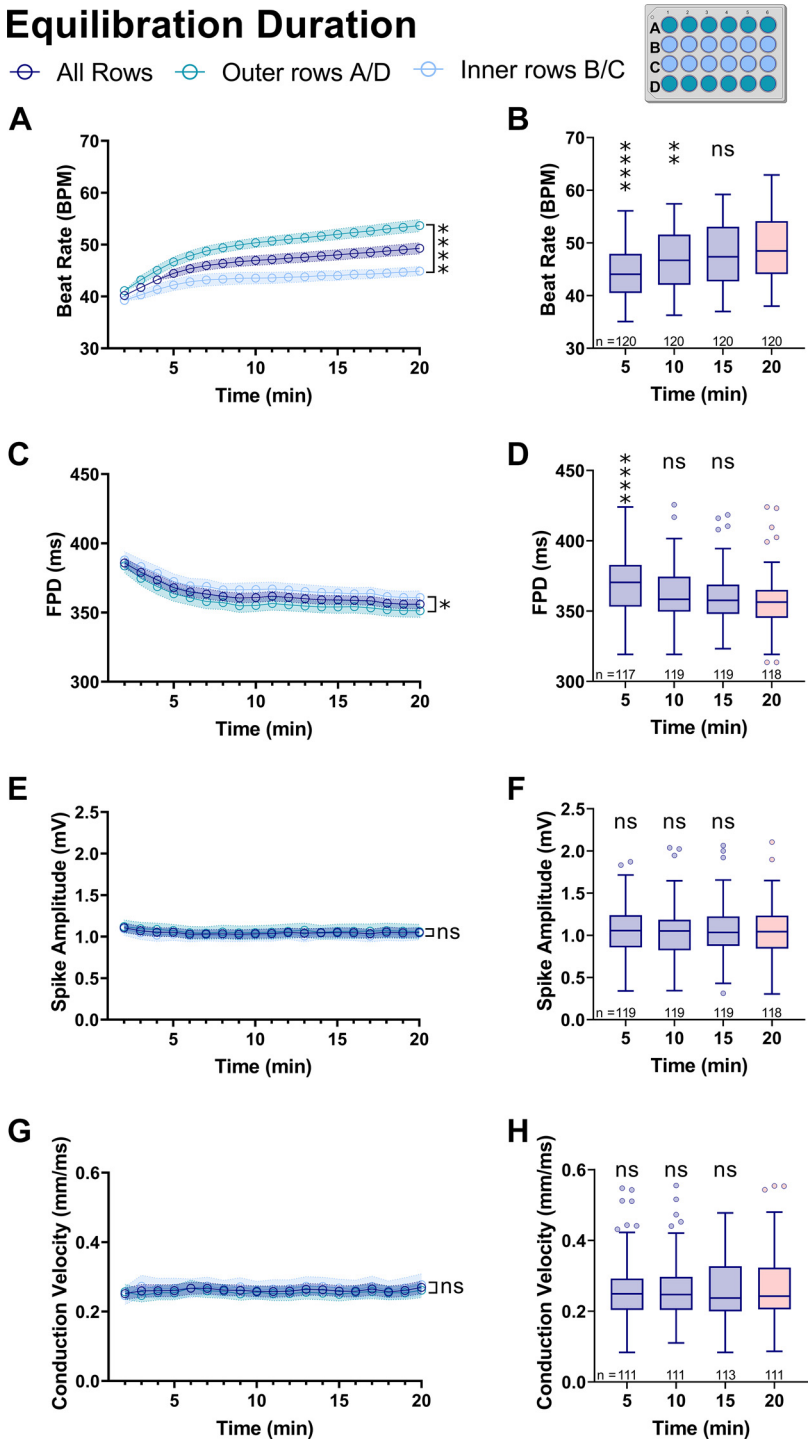


Figure 2. iCell Cardiomyocytes^2 equilibration duration. iCell Cardiomyocytes^2 electrophysiology data were collected continuously for 20 min (4 days post-defrost) during spontaneous beating. *A* and *B*: spontaneous beating rate (bpm, beats/min). *C* and *D*: field potential duration (FPD). *E* and *F*: depolarization spike amplitude. *G* and *H*: conduction velocity. *Left*: row location measurements are reported as the average of all wells across the MEA plate (purple), outer rows (green), or inner rows (blue). Open circle, mean, shaded region = 95% confidence interval. Statistical comparison was performed at 20-min time point (ANOVA with Holm–Sidak correction for multiple comparisons). All rows, n = 107–120; outer rows A/D, n = 54–60; inner rows B/C, n = 51–60. *Right*: equilibration time measurements are for all wells shown as a Tukey box-and-whisker plot, with outliers indicated. Statistical comparisons at 5-min intervals (purple box plots) vs. the end of equilibration at 20-min time point (red box plot). ANOVA with Holm–Sidak correction for multiple comparisons was used. *P < 0.05, **P < 0.01, ****P < 0.0001; ns, not significant. Number of replicates is indicated for each time point. ANOVA, analysis of variance; MEA, microelectrode array.

changes in the results (repeated-measures design, Fig. 3). FPD measurements were slightly shorter when the current was increased from 20 to 50 μA (308.9 ± 12 to 304.9 ± 15.1 ms, 0.29 Cohen effect size, P < 0.001), but unchanged by adjustments in the pulse duration or voltage amplitude (Fig. 3, *A–C*). Spike amplitude and conduction velocity measurements were influenced by changes in current, stimulus duration, and voltage amplitude (Fig. 3, *D–I*). For example, a faster conduction velocity was noted when the pulse duration was lengthened from 400 to 1,000 μs (0.207 ± 0.02 to 0.215 ± 0.03 mm/ms, 0.31 Cohen effect size, P < 0.001). To avoid misinterpreting results that are influenced by electrical stimulation settings, we recommend using a predefined setting to pace hiPSC-CMs at multiple frequencies (1–3 Hz). Such an approach will allow for a direct comparison of rate-dependent electrophysiology metrics before and after drug/chemical treatment, particularly in cases where cell excitability or the intrinsic beating rate changes.

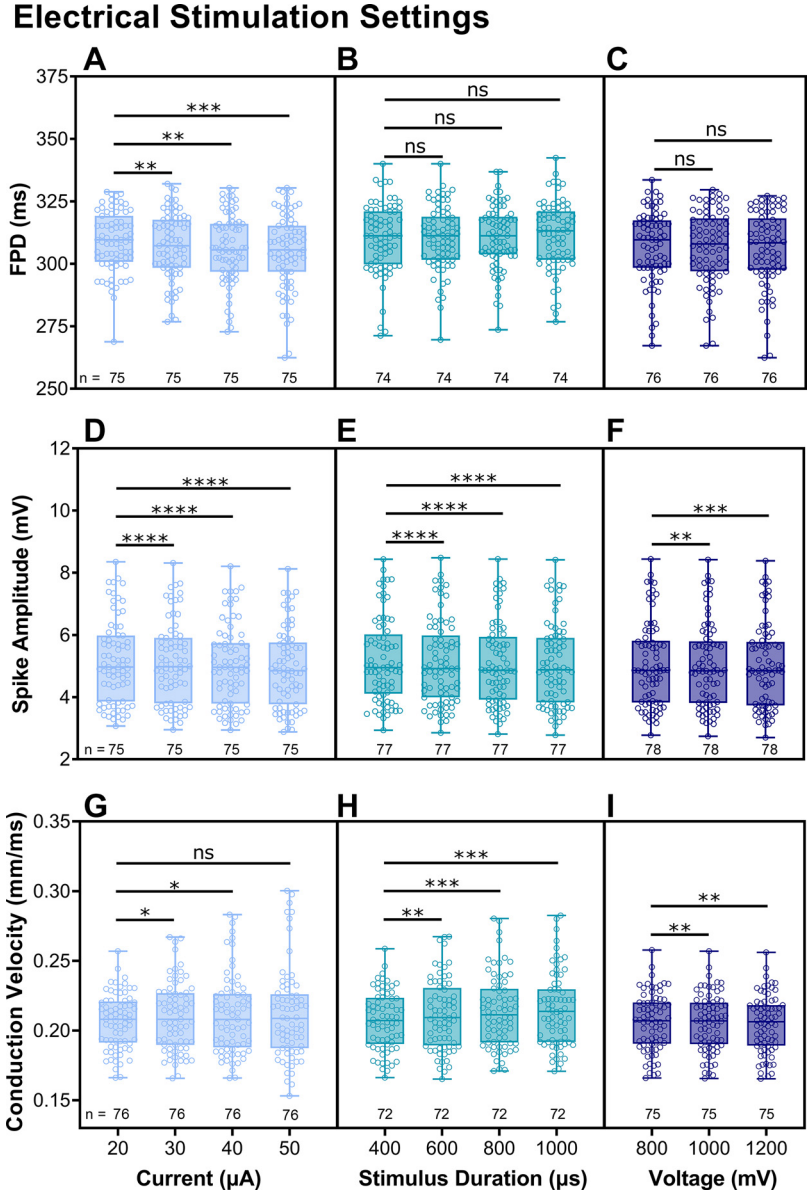


Figure 3. iCell Cardiomyocytes[2] electrical stimulation settings. Electrophysiology data were collected (7 days post-defrost) in response to external pacing at 1.5-Hz frequency (standard settings: 400-μs stimulus duration, 800-mV voltage amplitude, and 20-μA current). Each parameter was adjusted independently, with statistical comparison vs. the standard default setting. *Top*: changes in field potential duration (FPD) measurement following adjustment in applied current (*A*), stimulus duration (*B*), voltage amplitude (*C*). *Middle*: changes in spike amplitude following adjustment in applied current (*D*), stimulus duration (*E*), voltage amplitude (*F*). *Bottom*: changes in conduction velocity following adjustment in applied current (*G*), stimulus duration (*H*), and voltage amplitude (*I*). Measurements are shown as a box-and-whisker plot (all values shown, minimum to maximum). Statistical comparison vs. default standard setting. ANOVA with Holm–Sidak correction for multiple comparisons was used. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

Changes in hiPSC-CM Electrophysiology during Cell Culture

Previous studies have demonstrated that stem cell-derived cardiomyocytes continue to mature with longer cell culture time, which can directly impact cardiomyocyte electrophysiology (18, 57–59). Preclinical drug testing studies frequently use commercially available hiPSC-CM cell lines within the first 2 wk post-defrost. Therefore, we characterized daily phenotypic changes in iCell Cardiomyocytes[2] within this time frame while cells were spontaneously beating (Fig. 4, Supplemental Table S2). The spontaneous beating rate decreased 23.6% within the first 14 days of cell culture [2 days, 53.8 ± 5.1 (1.15 Cohen effect size); 7 days, 46.2 ± 7.8; 14 days, 41.1 ± 4.5 beats/min (2.64 Cohen effect size); $P < 0.0001$; Fig. 4*A*]. Per the manufacturer's instructions, the "optimal" day for hiPSC-CM experimentation is 7 days post-defrost (51). When compared with this 7-day time point, the beating rate was significantly different 2–5 days and 9–14 days post-defrost ($P < 0.001$), only 6 and 8 days post-defrost were comparable (Fig. 4*A*). We also observed a 1,100% increase in the depolarization spike amplitude and 90% lengthening of FPD with longer cell culture time, which was exaggerated during the 1st wk and plateaued slightly during the 2nd wk post-defrost (Fig. 4, *C–F*). For example, at 2 days post-defrost, the mean FPD was 288.9 ± 15.7 ms, which increased to 450.6 ± 39.5 ms at 7 days (5.38 Cohen effect size) and 545.9 ± 24.2 ms at 14 days (12.60 Cohen effect size). The depolarization spike amplitude started at 0.30 ± 0.24 mV at 2 days post-defrost and increased drastically to 2.45 ± 0.76 mV at 7 days (3.82 Cohen effect size) and 3.61 ± 0.77 mV at 14 days (5.80 Cohen effect size). Conduction velocity exhibited a sinusoidal trend across the first 2 wk of culture with an initial decrease in conduction velocity from 2 to 5 days post-defrost (2 days, 0.33 ± 0.09; 5 days, 0.21 ± 0.05 mm/ms), followed by a gradual increase (7 days, 0.26 ± 0.06 mm/ms),

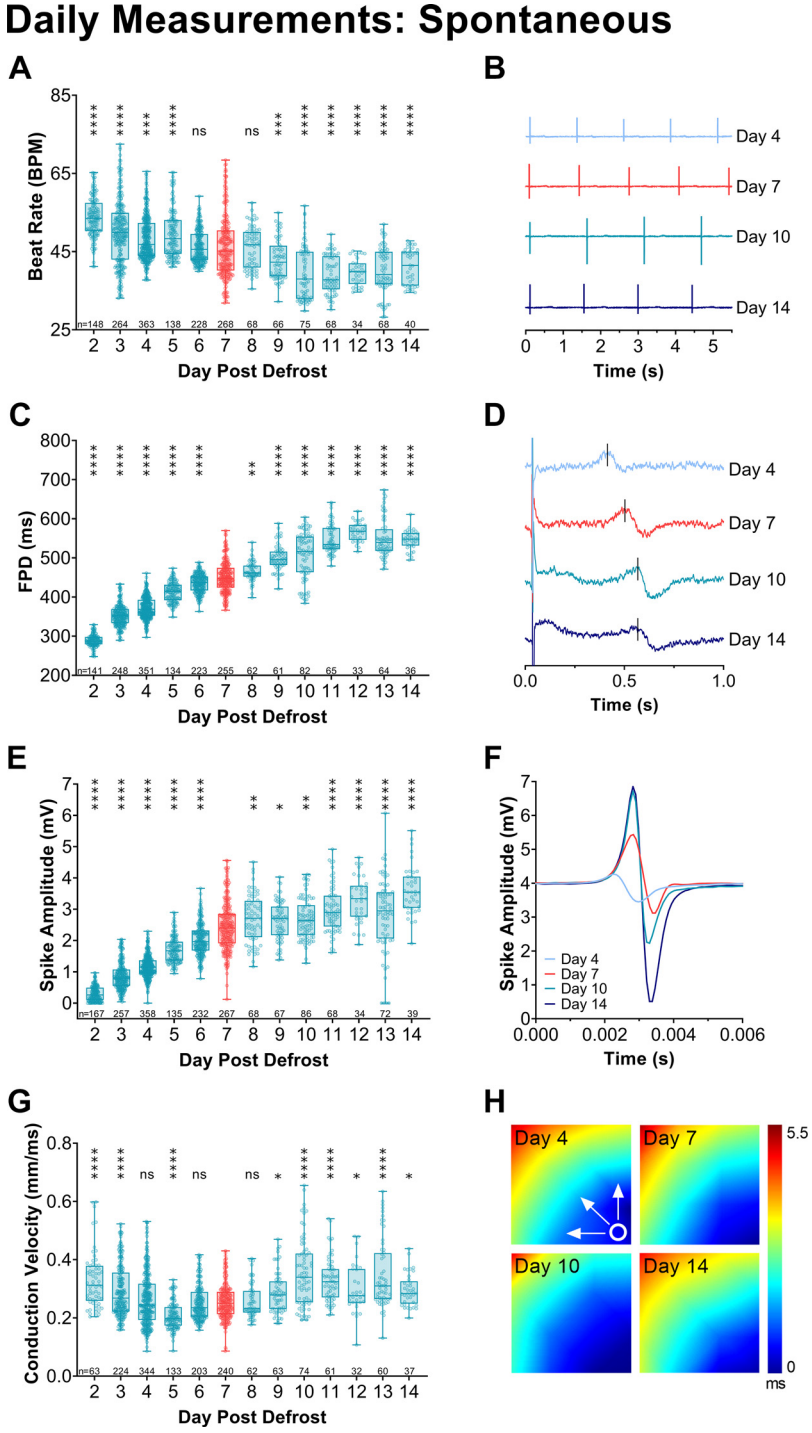


Figure 4. Daily changes in iCell Cardiomyocytes[2] electrophysiology measurements (spontaneously beating). *A–F*: during 2 wk of cell culture, spontaneous beating rate (bpm, beats/min) slows (*A* and *B*), field potential duration (FPD) lengthens (*C* and *D*), and depolarization spike amplitude increases (*E* and *F*). *G*: conduction velocity changed in a sinusoidal manner during 2 wk of cell culture. *H*: activation maps show electrical signal propagating across a cardiomyocyte layer at different time points. *Left*: daily measurements are shown as a box-and-whisker plot (all values shown, minimum to maximum), with statistical comparison vs. *day 7* values (shown in red). *Right*: representative traces at different time points. ANOVA with Holm–Sidak correction for multiple comparisons was used. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

and then another decline from 10 to 14 days (10 days, 0.35 ± 0.12 mm/ms; 14 days, 0.30 ± 0.06 mm/ms; Fig. 4, *G* and *H*).

Since the hiPSC-CM spontaneous beating rate slowed significantly during 2 wk of cell culture and cardiac electrophysiology metrics are rate dependent, we measured each variable again in response to external pacing (1.5 Hz). In response to pacing, FPD measurements were longer in iCell Cardiomyocytes[2] that were cultured for longer periods of time, although the magnitude of this change (21.9%; 3 days, 275.5 ± 13.8; 14 days, 336.0 ± 14 ms; 4.35 Cohen effect size; $P < 0.0001$) was less pronounced than that observed during spontaneous beating (54.8% 3 days vs. 14 days). Changes in paced FPD values were exaggerated during the 1st wk of culture, and then plateaued after 9 days in culture (Fig. 5*A*). In response to pacing, the depolarization spike amplitude also increased with longer cell culture time (81.6%; 3 days, 3.8 ± 1.1; 14 days, 6.9 ± 0.9 mV; 3.08 Cohen effect size; $P < 0.0001$; Fig. 5*B*), but again the magnitude was less pronounced than observed during spontaneous beating (319.9%; 3 days vs. 14 days). The

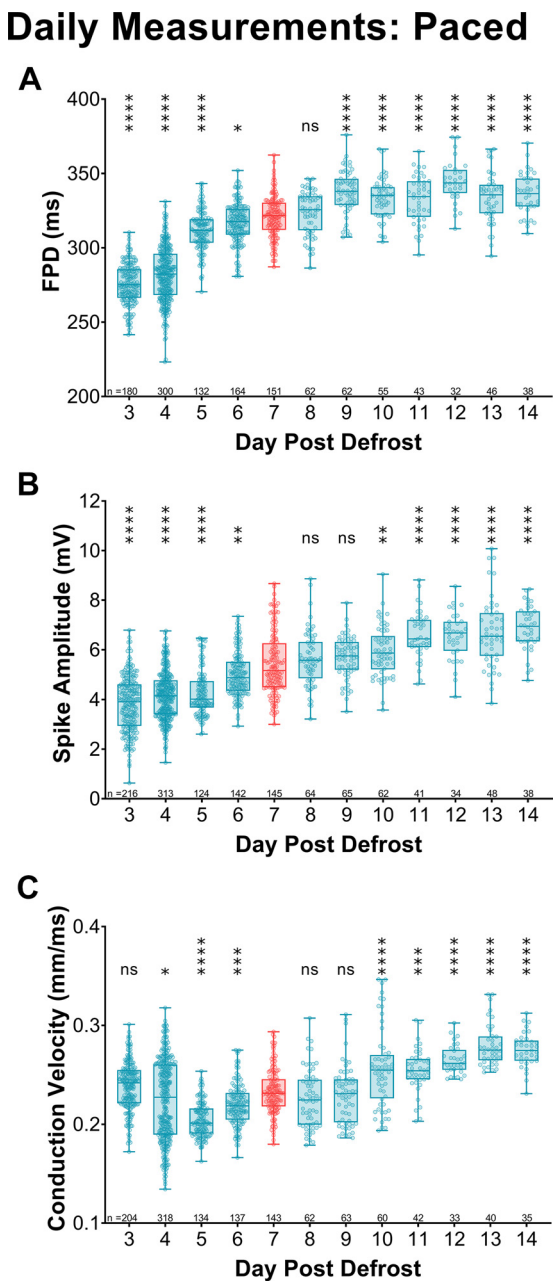


Figure 5. Daily changes in iCell Cardiomyocytes^2 electrophysiology measurements (external pacing, 1.5 Hz). *A*–*C*: during 2 wk of cell culture, field potential duration (FPD) lengthens (*A*), depolarization spike amplitude increases (*B*), and conduction velocity changes in a sinusoidal manner (*C*). Daily measurements are shown as a box-and-whisker plot (all values shown, minimum to maximum), with statistical comparison vs. *day 7* values (shown in red). ANOVA with Holm–Sidak correction for multiple comparisons was used. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, and ****$P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

depolarization spike amplitude was significantly larger when cells were paced versus spontaneously beating, for every time point investigated (e.g., 7 days spontaneous, 2.5 ± 0.8 mV; 7 days paced, 5.4 ± 1.3 mV; 2.69 Cohen effect size). Paced conduction velocity measurements exhibited a sinusoidal trend across the first 2 wk of culture, initially decreasing from 3 to 5 days post-defrost (3 days, 0.24 ± 0.02; 5 days, 0.20 ± 0.02 mm/ms; 2.0 Cohen effect size), then increasing during the remainder of the study period (7 days, 0.23 ± 0.02; 14 days, 0.28 ± 0.02 mm/ms; Fig. 5*C*). For reference, daily measurements are available in Supplemental Table S3.

MEA technology allows for the measurement of both extracellular field potentials (akin to the electrocardiographic QT interval) and local extracellular action potentials (akin to a monophasic action potential). It has previously been documented that the FPD measurement strongly correlates with an APD between 50 and 90% repolarization (28, 60). Similar to our FPD measurements, we found that the APD at 30% repolarization lengthened by 18.6% across multiple days in cell culture (4 days, 171.7 ± 19.4; 10 days, 203.7 ± 26.0 ms; 1.40 Cohen effect size; $P < 0.01$; Fig. 6*A*). APD lengthening was more dramatic at later phases of repolarization, increasing by 20.5% at APD_{50} (4 days, 247.6 ± 16.4; 10 days, 298.4 ± 20.5 ms; 2.74 Cohen effect size; $P < 0.01$; Fig. 6*B*) and 24.9% at APD_{90} (4 days, 319.6 ± 15.3; 10 days, 399.2 ± 15.5 ms; 5.17 Cohen effect size; $P < 0.01$; Fig. 6*C*). When compared with the "optimal" day for iCell Cardiomyocytes^2 experimentation [7 days post-defrost (51)], only 6–8 days post-defrost produced comparable APD_{90} values. For reference, daily measurements are available in Supplemental Table S4. We recommend using hiPSC-CM within a narrow time frame after defrosting (e.g., 6–8 days) and reporting the raw values for each cardiac metric to facilitate data interpretation and comparison between research laboratories.

Stability of hiPSC-CM Contractility Parameters during Cell Culture

Previous studies have demonstrated that hiPSC-CMs continue to mature in cell culture. Younger hiPSC-CMs have longer contractile events with smaller calcium transient amplitudes, while older cells display faster contraction, larger calcium transient amplitudes, and prolonged relaxation (4 wk) (58, 59). In the current study, we only observed slight differences in excitation-contraction delay and beat amplitude, and these adaptations were limited to the first few days of cell culture as the monolayer was forming (Fig. 7). The time delay between excitation and contraction lengthened from *day 2* to *day 4* (2 days, 248.8 ± 30.6; 4 days, 277.5 ± 29.5; 0.95 Cohen effect size; $P < 0.0001$; Fig. 7*A*) and then plateaued, but only during spontaneous beating. The beat amplitude decreased in the first few days of cell culture, during both spontaneous (2 days, 2.2 ± 0.4; 5 days, 1.7 ± 0.3%; 1.41 Cohen effect size) and paced (3 days, 1.6 ± 0.3; 5 days: 1.4 ± 0.3%; 0.67 Cohen effect size) recordings (Fig. 7, *B* and *D*). Since these experiments were only performed up to 14 days in culture, it is possible that additional adaptations occur at later time points.

Changes in hiPSC-CM Drug Responsiveness during Cell Culture

Previous studies have reported profound differences in cardiac gene expression and key ionic currents in stem cell-derived cardiomyocytes after long-term cell culture (weeks to months) (18, 58, 59) and/or the implementation of maturation-specific protocols (16–21). We investigated whether such

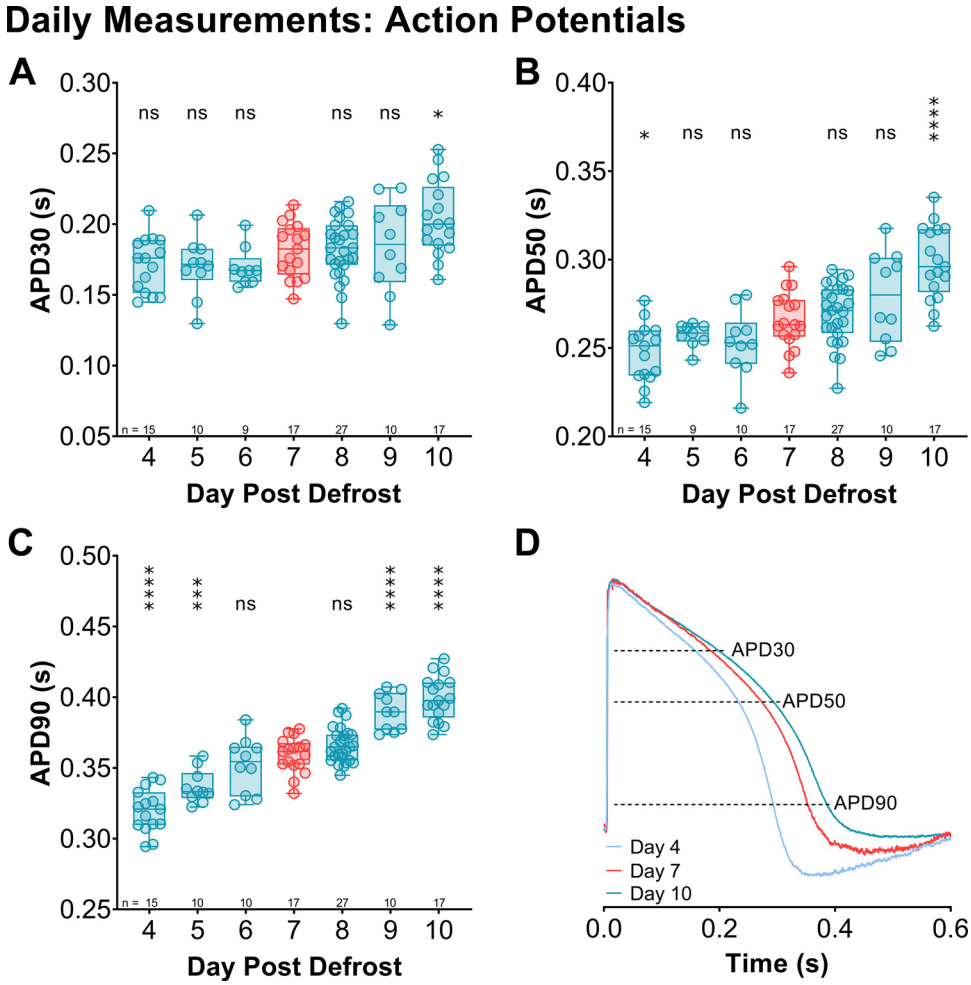


Figure 6. Daily changes in iCell Cardiomyocytes[2] action potential duration (external pacing, 1.5 Hz). *A*–*C*: action potential duration at 30% repolarization (APD_{30}; *A*), 50% repolarization (APD_{50}; *B*), 90% repolarization (APD_{90}; *C*). *D*: representative example of action potential recordings. Daily measurements are shown as a box-and-whisker plot (all values shown, minimum to maximum), with statistical comparison vs. *day 7* values (shown in red). ANOVA with Holm–Sidak correction for multiple comparisons. $*P < 0.05$, $***P < 0.001$, and $****P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

underlying developmental differences could alter drug responsiveness, even within the relatively short time frame recommended for hiPSC-CM experimentation (51). First, we observed that the iCell Cardiomyocyte[2] spontaneous beating rate was altered by the application of a β-adrenergic agonist (isoproterenol), calcium channel blocker (nifedipine), or hERG blocker (E-4031), but the magnitude of this change was influenced by cell culture duration (Fig. 8*A*). Isoproterenol caused a slight increase in the beating rate (11.5%) on *day 4* post-defrost, which was more pronounced (48.8%) on *day 10*. Conversely, nifedipine increased the beating rate 112.9% on *day 4* post-defrost, but its effect was more tempered (56.8%) on *day 10*. E-4031 caused a slight increase in the beating rate (4.5%) on *day 4* post-defrost, while significant beat rate slowing was observed (−19.1%) on *day 10* post-defrost. Second, the application of E-4031 lengthened the FPD (4 days, 31.2%; 10 days, 17.2%) and APD_{90} (4 days, 41.5%; 10 days, 10.0%; Fig. 8, *C* and *H*) but the effects were more exaggerated at the earlier time point. Third, the application of isoproterenol increased conduction velocity at the later time point (4 days, −6.5%; 10 days, 3.5%) and the depolarization spike amplitude was also more affected at the later time point (4 days, −12.3%; 10 days, −27.3%; Fig. 8, *D* and *E*). The length of cell culture time should be considered when performing drug screening or safety testing, as hiPSC-CMs may respond differently depending on their maturation state.

Validation of Key Results Using an Alternative hiPSC-CM Cell Line

To determine whether daily adaptations in iCell Cardiomyocytes[2] were cell line specific, we repeated the described experiments using Cor.4U cardiomyocytes—a cell line previously used for comparison studies (31, 32, 61, 62). We noted similar daily trends in Cor.4U cells cultured for 2–14 days. When spontaneously beating, we observed lengthening of FPD (*day 2*, 182.0 ± 13.3; *day 14*, 236.5 ± 22.8 ms; 2.92 Cohen effect size) and a larger spike amplitude (*day 2*, 0.23 ± 0.15; *day 14*, 8.09 ± 3.09 mV; 3.59 Cohen effect size; Supplemental Fig. S3 and Supplemental Table S5). A few notable differences included a relatively stable spontaneous beating rate during the 2-wk cell culture period (*day 2*, 73.0 ± 4.68; *day 14*, 71.6 ± 7.9 beats/min; 0.22 Cohen effect size) and a steadily increasing conduction velocity (*day 2*, 0.072 ± 0.016; *day 14*, 0.212 ± 0.039 mm/ms; 4.70 Cohen effect size). Since Cor.4U cells have a faster intrinsic beating rate (32), a slightly faster pacing frequency (2 Hz) was used for subsequent experiments (compared with 1.5 Hz for iCell cardiomyocytes[2]). Again, we noted similar trends in Cor.4U cells cultured for 3–14 days—including a larger spike amplitude (*day 3*, 0.72 ± 0.17; *day 14*, 8.38 ± 3.38 mV; 3.20 Cohen effect size) and a faster conduction velocity (*day 3*, 0.07 ± 0.01; *day 14*, 0.20 ± 0.04 mm/ms; 16.79 Cohen effect size; Supplemental Fig. S4 and Supplemental

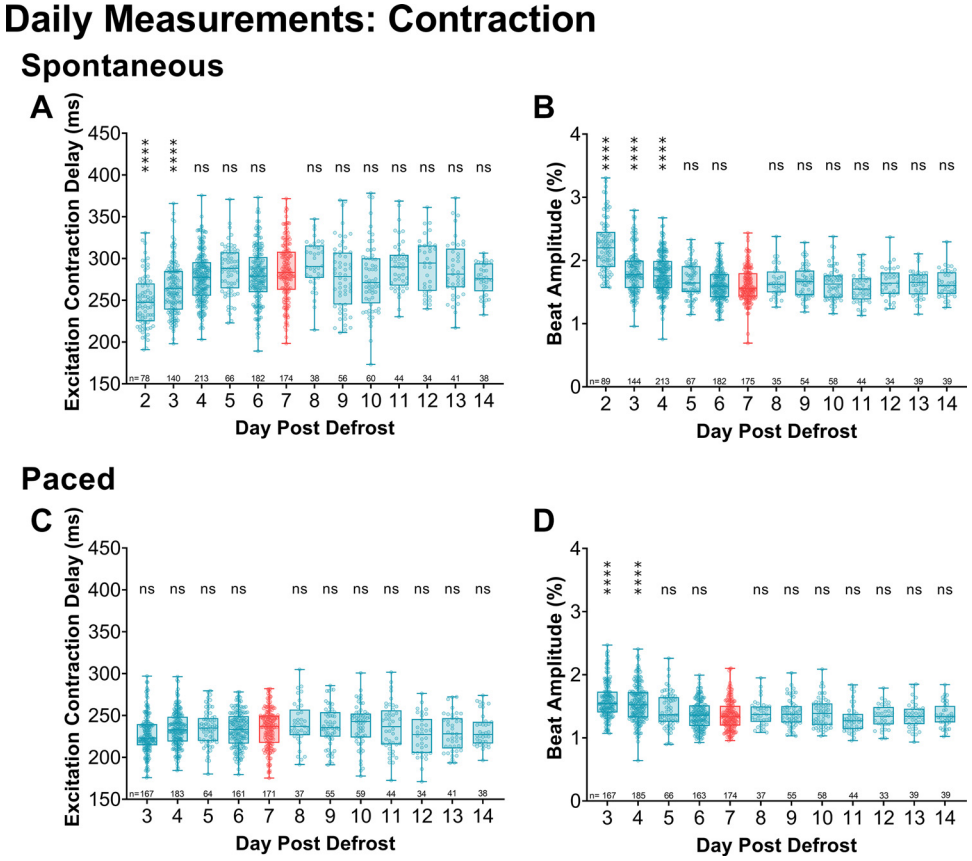


Figure 7. Daily changes in iCell Cardiomyocytes[2] contraction measurements. *A* and *C*: daily changes in excitation-contraction delay. *B* and *D*: daily changes in beat amplitude. *Top*: measurements collected daily during 2 wk of cell culture, while cells were spontaneously beating. *Bottom*: measurements in response to external pacing (1.5 Hz). Daily measurements are shown as a box-and-whisker plot (all values shown, minimum to maximum), with statistical comparison vs. *day 7* values (shown in red). ANOVA with Holm–Sidak correction for multiple comparisons. ****$P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

Table S6). One distinguishing feature was that Cor.4U cells had minimal changes (3 vs. 14 days) in the measured FPD, under paced conditions. Finally, we evaluated the length of time required for Cor.4U cells to reach a steady state equilibrium in the MEA system (on *day 7* post-defrost). Although Cor.4U cells had minor fluctuations during the 20-min equilibration time frame, none of the measured metrics (beating rate, FPD, spike amplitude, and conduction velocity) were significantly different between time points (5-, 10-, and 15-min equilibration; Supplemental Fig. S5 and Supplemental Table S7). Prior experimental work on neonatal rat cardiomyocytes suggest that temperature fluctuations have less of an impact on the spontaneous beating rate of more mature cardiomyocytes (63). Accordingly, we speculated that Cor.4U cells may be more developmentally mature, as compared with iCell cardiomyocytes[2].

hiPSC-CM Gene Expression Profile Changes during Cell Culture

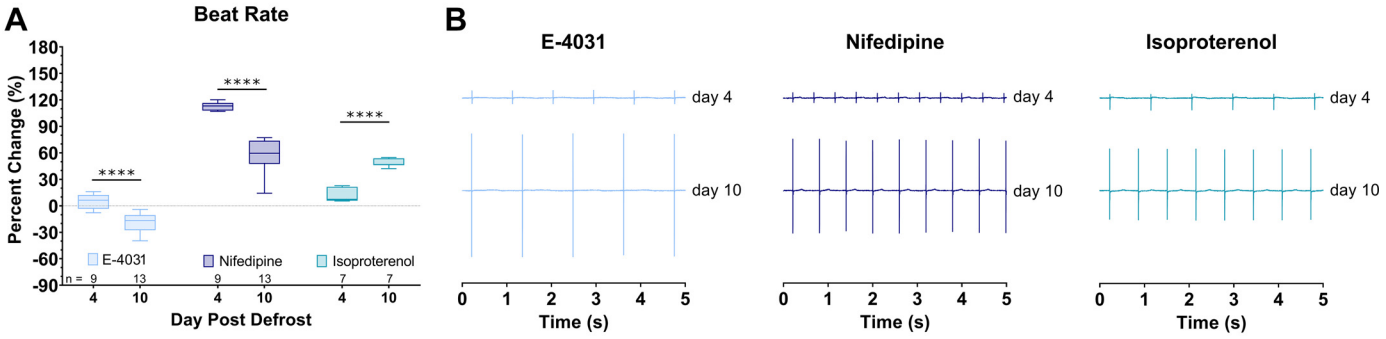
Nozaki et al. previously reported that the gene expression profile of hiPSC-CMs changes within the first 2 wk of cell culture (35). Accordingly, we tested whether daily adaptations in cardiac electrophysiology measurements might be linked to underlying changes in hiPSC-CM maturity. Between 2 and 30 days in cell culture, we observed an age-dependent increase in the expression of key cardiac genes in iCell cardiomyocytes[2], including 2.3-fold higher *TNNI3* (cardiac type, troponin I3), 2.3-fold higher *CACNA1C* (voltage-gated L-type calcium channel), 8.1-fold higher *CASQ2* (calsequestrin-2), 3.1-fold higher *SCN5A* (voltage-gated sodium channel), and 27.5-fold higher *KCNJ2* (inwardly rectifying potassium channel; Fig. 9). We also confirmed that cardiomyocyte purity was maintained, as *VIM* (vimentin, a fibroblast marker) expression did not increase during the 30-day cell culture period. In comparison, the gene expression profile of Cor.4U cells shifted, but to less of an extent, between 2 and 30 days in cell culture. And although *KCNJ2* expression also increased over time, the change was to a lesser magnitude (5.0-fold Cor.4U vs. 27.5-fold iCell cardiomyocytes[2], 2–30 days) suggesting that Cor.4U cells may have a more mature phenotype at the time of defrost.

DISCUSSION

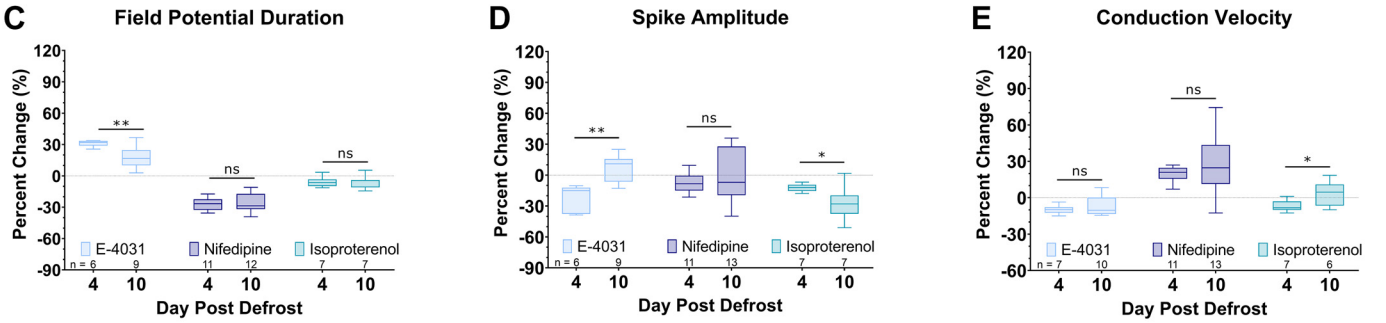
In the presented study, we used a commercially available hiPSC-CM cell line (iCell Cardiomyocytes[2], Fujifilm Cellular Dynamics) that has been widely used to test the efficacy, safety, and toxicity of pharmacological agents and environmental chemicals (48, 55, 61, 64–66). To the best of our knowledge, this is the first study to monitor daily changes in hiPSC-CM physiology throughout 14 days in cell culture, which is within the manufacturer's recommended time frame for experimentation. We then used an alternative cell line (Cor.4U cardiomyocytes, Axiogenesis/Ncardia) to assess whether the observed trends were consistent across cell lines. Our study observed multiple sources of variability that impacted hiPSC-CM MEA measurements, including *1*) well-to-well variability across the MEA plate, *2*) length of

Drug Responsiveness

Spontaneous Electrophysiology



Paced Electrophysiology



Paced Action Potential

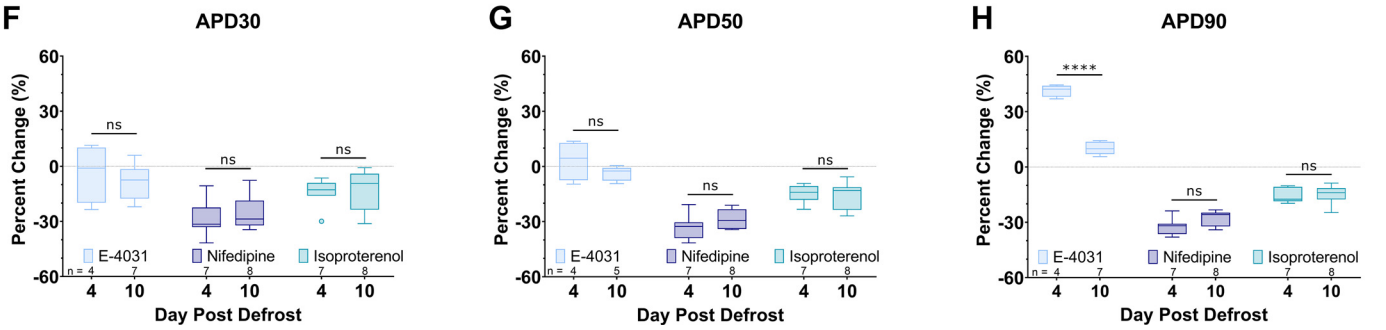


Figure 8. iCell Cardiomyocytes^2 drug responsiveness changes based on the length of cell culture time. Electrophysiology data were collected (4 vs. 10 days post-defrost) at baseline and after exposure to a hERG channel blocker (E-4031), calcium channel blocker (nifedipine), or β-adrenergic agonist (isoproterenol). *A*: percent change in the spontaneous beating rate, normalized to control after a 15-min drug treatment. *B*: representative traces of spontaneous beating after drug treatment. *C–E*: percent change in the field potential duration (*C*), spike amplitude (*D*), and conduction velocity (*E*; normalized to control measurements) after 15 min of drug treatment. *F–H*: percent change in the action potential duration at 30% (*F*), 50% (*G*), and 90% (*H*) repolarization (normalized to control measurements) after 15 min of drug treatment. For paced recordings, iCell Cardiomyocytes^2 were externally paced to 1.5 Hz for E-4031, which slowed the intrinsic rate, and 3 Hz for nifedipine and isoproterenol, which increased the intrinsic rate. Measurements are shown as a Tukey box-and-whisker plot. Unpaired Student's *t* test, two tailed. $^*P < 0.05$, $^{**}P < 0.01$, and $^{****}P < 0.0001$; ns, not significant. Number of replicates indicated.

equilibration period as hiPSC-CMs reach a steady state within the MEA system, *3*) electrical stimulation settings used to externally pace hiPSC-CMs, *4*) duration of post-defrost cell culture, and *5*) the type of hiPSC-CM cell line. Collectively, our study highlights important user-defined experimental factors that should be considered when using hiPSC-CM models for disease modeling, safety pharmacology, and cardiotoxicity screening.

In our study, we were surprised to find that the well location within an MEA plate impacted baseline electrophysiology measurements. Specifically, iCell Cardiomyocytes^2 plated in the top/bottom rows had faster beating rates and shorter FPD values compared with those plated in the inner rows. These results highlight an important consideration, as the location of treatment groups on an MEA plate could give the (mis)perception of drug effects (e.g., vehicle control treatment

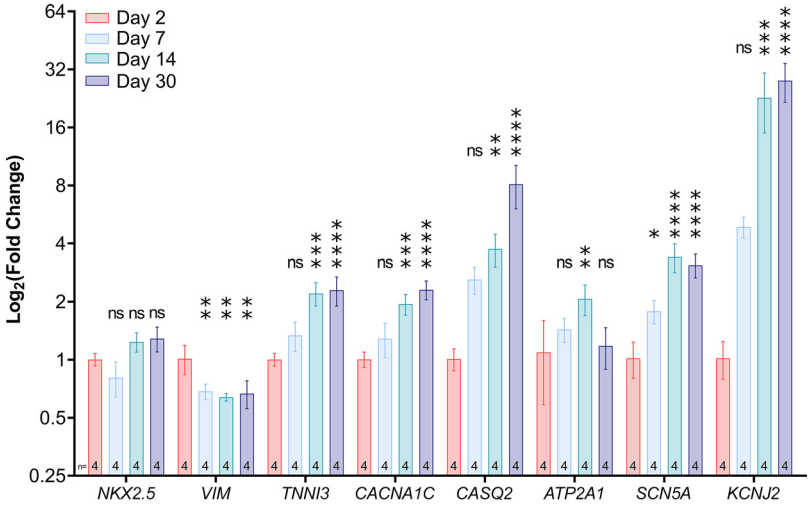


Figure 9. Changes in iCell Cardiomyocytes[2] gene expression across 1 mo of culture. Genes of interest included homeobox protein (*NKX2.5*; a marker of heart development), vimentin (*Vim*; a fibroblast marker), troponin I (*TNNI3*; a marker of cardiac muscle contraction), voltage-dependent L-type calcium channel α-1 C (*CACNA1C*, a marker of calcium influx), calsequestrin-2 (*CASQ2*, a marker of calcium storage), sarco(endo)plasmic reticulum calcium ATPase 1 (*ATP2A1*, a marker of calcium reuptake), sodium channel protein type 5 subunit-α (*SCN5A*, a marker of action potential propagation), and inward rectifier potassium channel 2 (*KCNJ2*, a marker of the resting membrane potential). Ordinary one-way ANOVA with multiple comparisons to *day 2* (shown in red) using Holm–Sidak correction was used. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$; ns, not significant. Number of replicates indicated. ANOVA, analysis of variance.

added only to the top row of cells, drug treatment in a middle row). Potential explanations for this variability may include slight differences in temperature distribution across the plate. Unfortunately, we were not able to pinpoint the specific cause of these differences, since measuring the cell culture media temperature required opening the MEA system (which in turn, further disturbs the atmosphere). Well-to-well variability persisted even after allowing iCell Cardiomyocytes[2] to equilibrate to the new environment within the MEA system for 20 min. Furthermore, we noted a progressive increase in beating rate and shortening of FPD during this equilibration period. Accordingly, even slight modifications in equilibration time between studies can influence MEA data and may contribute to interfacility variability. For example, Nozaki et al. reported that hiPSC-CM MEA studies accurately predict the risk of drug-induced arrhythmias, but the minimum concentration required to produce arrhythmia-like waveforms can vary by 10-fold when results are compared between multiple laboratories (despite using the same hiPSC-CM cell line and batch) (34). Kitaguchi et al. (67) also documented intrafacility and interfacility variability in basal hiPSC-CM measurements and drug response and reported that the drug concentration inducing FPD prolongation ranged from 1.8 to 5.8-fold between different laboratories using the same cell line. As another example, Blinova et al. (30) reported interfacility variability between 10 different laboratories (0.69–0.88 cross-site correlation) in detecting drug-induced repolarization prolongation in 28 tested drugs. Potential sources of such variability may include well-to-well differences, slight deviations in equilibration time, minor modifications in cell culture techniques (e.g., plating density, age of cells), or MEA recording parameters. Indeed, in this study, we found that minor adjustments in the pulse width, voltage amplitude, or applied current can alter cardiac electrophysiology measurements. To aid in data interpretation, hiPSC-CM MEA studies should report important technical details (e.g., pacing frequency, stimulation settings, cell culture duration) and consider reporting raw data values to facilitate data comparison between studies. To support reproducibility and data interpretation, we have included a detailed table that researchers can use to report key experimental conditions for future hiPSC-CM studies that use MEA recordings (see Supplemental Table S8).

Another critical parameter to consider when performing MEA experiments is the length of time hiPSC-CMs are grown in culture. hiPSC-CMs are immature cardiomyocytes with a fetal-like phenotype that are in a continued state of maturation (for review, see Refs. 14, 68). As such, extended periods of cell culture have been shown to shift the phenotype of these cells to a more adult-like state (18, 58, 69). Between 1 and 4 wk of cell culture, Kumar et al. (59) reported that older hiPSC-CMs have a more mature ultrastructure with increased myofibril organization, refined sarcomere structures, developed intercalated disks, and increased mitochondrial density. Older 4-wk hiPSC-CMs also have increased expression for genes encoding cardiac ion channels (e.g., *KCNJ2*, *KCNQ1*, *SCN5A*) and calcium handling proteins (e.g., *RYR2*, *CASQ2*). In agreement, Nozaki et al. (35) reported a stark shift in cardiac gene expression within the short time frame (1–15 days) recommended for MEA studies using a commercially available hiPSC-CM cell line. The latter included a 12–47-fold difference in key genes associated with voltage-gated ion channel activity (e.g., *KCNJ1*, *KCNQ1*, *CACNA2D2*) and muscle contraction (e.g., *CASQ2*, *MYBPC3*). Similarly, our studies suggest that hiPSC-CMs continue to mature post-defrost and that this process may shift the phenotype of these cells. Specifically, we noted an increase in genes related to intracellular calcium handling (*CACNA1C*, *CASQ2*), electrophysiology (*SCN5A*, *KCNJ2*), and contractile machinery (*TNNI3*) in iCell cardiomyocytes[2] cultured for longer periods of time. We also observed a steady increase in *KCNJ2* expression in Cor.4U cells with prolonged culture time, albeit to a lesser extent. Previous studies have documented intrinsic variations in the cardiac gene expression profiles of Cor.4U and iCell Cardiomyocytes (not iCell Cardiomyocytes[2]), and in both studies, cell samples were collected and pooled across a range of cell culture time points [4–14 days (62) and 7–17 days (31)].

Collectively, these results suggest that passive maturation in cell culture is likely to influence hiPSC-CM electrophysiology under baseline conditions and in response to various drug compounds, which could also account for interfacility

variability in MEA studies. In the current 2-wk-long study, we reported significant slowing of iCell Cardiomyocytes^2 beating rate, increased depolarization spike amplitude, and lengthening of FPD and APD under standard cell culture conditions. Similarly, we observed increased depolarization spike amplitude, lengthened FPD, and faster conduction velocity in spontaneously beating Cor.4U cardiomyocytes during the 2-wk-long study. These findings agree with the study by Kumar et al. (59), which reported that extended cell culture time (4 wk) slowed the beating rate and lengthened FPD in hiPSC-CMs, and Doss et al. (70) which reported longer APD_{90} in older hiPSC-CM (~4 mo). In the current study, our analysis of contractile function was limited to excitation-contraction coupling delay time and beat amplitude measurements. Even though these parameters were not changed during this 2-wk study, others have reported on the maturation of intracellular calcium handling within 15–30 days of culture (71) and larger calcium current density within 30–80 days of cell culture (58). Since we observed daily adaptations in hiPSC-CM physiology, it is beneficial to use these cells within a well-defined time range, and long-term cardiotoxicity studies may require additional planning.

Notably, we also observed that iCell Cardiomyocyte^2 drug responsiveness is dynamic within the first 2 wk of cell culture. As an example, younger hiPSC-CMs (4 days) were more responsive to E-4031 (I_{Kr} blocker) as compared with slightly older hiPSC-CMs (10 days of culture), as demonstrated by a more significant lengthening of FPD and APD. In agreement, Da Rocha et al. (57) reported that immature hiPSC-CMs with a "fetal-like" phenotype had an exaggerated response to E-4031 with longer APDs and an increased incidence of early delayed afterdepolarizations, as compared with "adult-like" hiPSC-CMs that underwent an extracellular matrix-mediated maturation protocol. This variable drug response may be attributed to the reliance of immature hiPSC-CMs on I_{Kr} for repolarization, as compared with older hiPSC-CMs (70). We also observed that older hiPSC-CMs (10 days) were more responsive to isoproterenol than younger hiPSC-CMs (4 days of culture), as demonstrated by a larger increase in the spontaneous beating rate and faster conduction velocity. This variable isoproterenol response could be related to maturation-dependent changes in both β-adrenergic receptor expression and cAMP signaling (72, 73).

To improve preclinical cardiotoxicity testing, hiPSC-CM MEA experiments have been adopted as a vital component of the CiPA initiative. Implementing and rigorously reporting experimental techniques can improve reproducibility both within and between laboratories, and further support the utility of hiPSC-CMs as a high throughput screening tool. Toward this goal, our study highlights a few parameters to control for when planning hiPSC-CM MEA studies, including cell age, type of hiPSC-CM cell line, equilibrium duration, layout of drug treatments across a multiwell plate, and pacing stimulus settings. To date, the cardiac research field largely reports MEA measurements as a percent change from baseline, but also reporting raw data values will help to improve data interpretation between published studies.

A few limitations in our study design should be considered and expanded upon by future work. First, the current study was limited to the use of hiPSC-CMs that are a mixture of cardiomyocytes with ventricular, atrial, and nodal-like attributes (7, 74, 75). Both our current study and prior studies (27, 29, 31, 32) have documented phenotypic differences between cell lines (e.g., Cor.4U cardiomyocytes have a faster beating rate and shorter FPD compared with iCell Cardiomyocytes). Accordingly, the results of our study may not directly translate to other cell lines. Second, our study was limited to the use of a single MEA device; therefore, our results may/not be replicated in alternative high-throughput screening systems (e.g., microelectrode array, impedance based, optical based). Third, we monitored hiPSC-CMs physiology throughout the first few weeks post-defrost, which aligns with the manufacturer instructions on the appropriate time frame for experimentation. Nevertheless, hiPSC-CM maturation continues during prolonged cell culture, which can impact cardiac electrophysiology and contractility measurements at baseline and in response to drug or chemical exposure (58, 59, 76).

DATA AVAILABILITY

The authors confirm that the data supporting the findings of this study are available within the article and online supporting tables.

SUPPLEMENTAL DATA

Supplemental Figs. S1–S5 and Supplemental Tables S1–S8: https://doi.org/10.6084/m9.figshare.24222883.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

D.G. and N.P. conceived and designed research; D.G., J.P., S.S., and N.P. performed experiments; D.G., J.P., S.S., and N.P. analyzed data; D.G., J.P., S.S., and N.P. interpreted results of experiments; D.G., J.P., and N.P. prepared figures; D.G., J.P., S.S., and N.P. drafted manuscript; D.G. and N.P. edited and revised manuscript; D.G., J.P., S.S., and N.P. approved final version of manuscript.

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